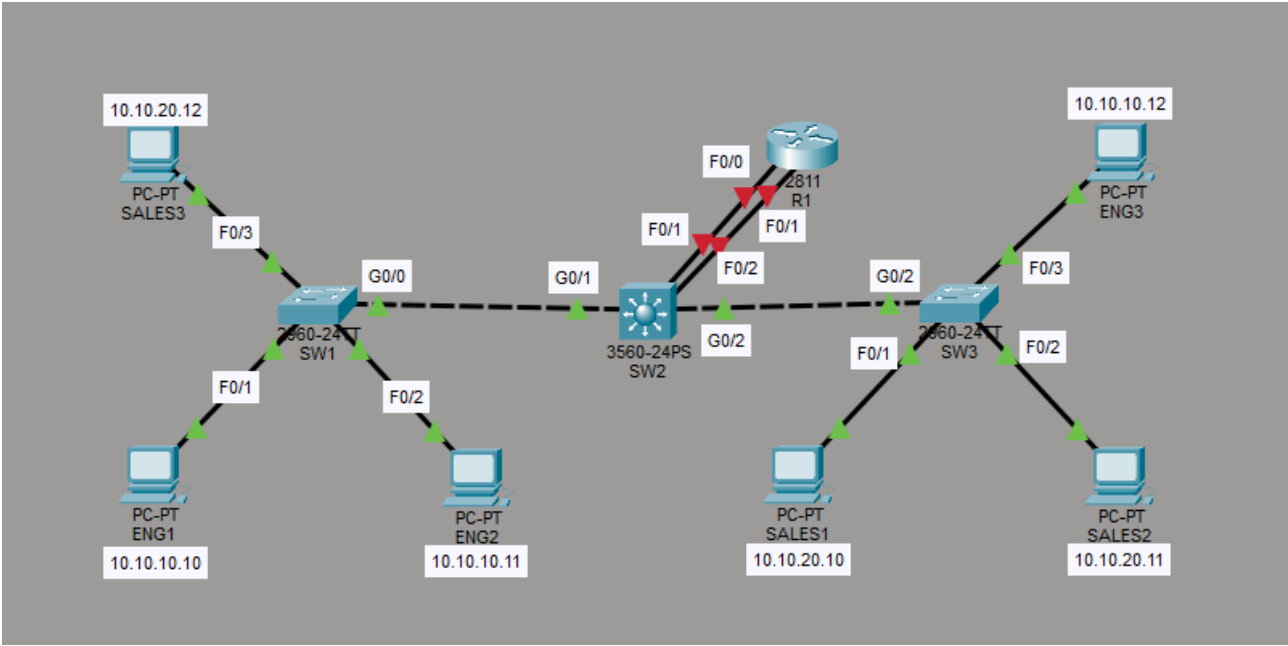


NET512 — Weekly Practical

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Topology



Task 89 — VTP Configuration

The image displays three screenshots of Cisco IOS Command Line Interfaces for switches SW2, SW1, and SW3, showing the configuration and status of VTP (VLAN Trunking Protocol).

SW2 Configuration:

```
SW2#
SW2(config)#vtp mode server
SW2(config)#vtp password pass123
SW2(config)#vtp version 2
SW2(config)#end
SW2#
*SYS-5-CONFIG_I: Configured from console by console

SW2#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : W9
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : 00E0.F71B.5B00
Configuration last modified by 0.0.0.0 at 3-1-93 00:08:24
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
-----
VTP Operating Mode       : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs : 5
Configuration Revision    : 1
MDS digest               : 0x31 0x09 0x5B 0x1D 0x02 0x3D 0xED 0x42
                          : 0xEF 0x46 0x91 0xCF 0x1C 0x25 0x43 0xCC

SW2#
```

SW1 Configuration:

```
SW1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#vtp domain W9
Domain name already set to W9.
SW1(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW1(config)#vtp password pass123
Setting device VLAN database password to pass123
SW1(config)#end
SW1#
*SYS-5-CONFIG_I: Configured from console by console

SW1#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : W9
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : 00D0.BC08.D600
Configuration last modified by 0.0.0.0 at 3-1-93 00:08:24

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
Configuration Revision    : 1
MDS digest               : 0x31 0x09 0x5B 0x1D 0x02 0x3D 0xED 0x42
                          : 0xEF 0x46 0x91 0xCF 0x1C 0x25 0x43 0xCC

SW1#
```

SW3 Configuration:

```
SW3#conf term
Enter configuration commands, one per line. End with CNTL/Z.
SW3(config)#vtp domain W9
Domain name already set to W9.
SW3(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW3(config)#vtp password pass123
Setting device VLAN database password to pass123
SW3(config)#end
SW3#
*SYS-5-CONFIG_I: Configured from console by console

SW3#show vtp status
VTP Version capable      : 1 to 2
VTP version running      : 2
VTP Domain Name          : W9
VTP Pruning Mode         : Disabled
VTP Traps Generation     : Disabled
Device ID                 : 0001.6433.1800
Configuration last modified by 0.0.0.0 at 3-1-93 00:08:24

Feature VLAN :
-----
VTP Operating Mode       : Client
Maximum VLANs supported locally : 255
Number of existing VLANs : 5
Configuration Revision    : 1
MDS digest               : 0x31 0x09 0x5B 0x1D 0x02 0x3D 0xED 0x42
                          : 0xEF 0x46 0x91 0xCF 0x1C 0x25 0x43 0xCC

SW3#
```

Task 90 — VLAN Creation

The image displays three screenshots of Cisco IOS Command Line Interface (CLI) windows, each showing the configuration of VLANs on a different switch (SW2, SW1, and SW3).

SW2 Configuration:

```
SW2(config-vlan)#name ENG
SW2(config-vlan)#exit
SW2(config)#vlan 20
SW2(config-vlan)#name SALES
SW2(config-vlan)#exit
SW2(config)#vlan 99
SW2(config-vlan)#name NATIVE
SW2(config-vlan)#exit
SW2(config)#end
SW2#
*SYS-5-CONFIG_I: Configured from console by console
SW2#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10 ENG	active	
20 SALES	active	
99 NATIVE	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

SW1 Configuration:

```
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
SW1(config-if)#description ENG2
SW1(config-if)#exit
SW1(config)#int fa0/3
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 20
SW1(config-if)#description SALES3
SW1(config-if)#end
SW1#
*SYS-5-CONFIG_I: Configured from console by console
SW1#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10 ENG	active	Fa0/1, Fa0/2
20 SALES	active	Fa0/3
99 NATIVE	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

SW3 Configuration:

```
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 20
SW3(config-if)#description SALES2
SW3(config-if)#exit
SW3(config)#int fa0/3
SW3(config-if)#switchport mode access
SW3(config-if)#switchport access vlan 10
SW3(config-if)#description ENG3
SW3(config-if)#end
SW3#
*SYS-5-CONFIG_I: Configured from console by console
SW3#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1
10 ENG	active	Fa0/3
20 SALES	active	Fa0/1, Fa0/2
99 NATIVE	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Task 91 — Trunk Ports

The image displays three screenshots of Cisco IOS Command Line Interfaces (CLI) for switches SW1, SW2, and SW3, showing the configuration and status of trunk ports.

SW1:

```
SW1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

SW1(config-if)#switchport trunk native vlan 99
SW1(config-if)#switchport trunk allowed vlan 10,20,99
SW1(config-if)#description SW
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (99), with SW2 GigabitEth
SW1(config-if)#description SW2 Trunk
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (99), with SW2 GigabitEthernet0/1 (1).

SW1#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    on        802.1q         trunking      99

Port      Vlans allowed on trunk
Gig0/1    10,20,99

Port      Vlans allowed and active in management domain
Gig0/1    none

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    none

SW1#
```

SW2:

```
SW2(config-if)#switchport trunk native vlan 99
SW2(config-if)#switchport trunk allowed vlan 10,20,99
SW2(config-if)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (99), with SW3 GigabitEthernet0/2 (1).

SW2#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/2 (99), with SW3 GigabitEthernet0/2 (1).

SW2#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/1    on        802.1q         trunking      99
Gig0/2    on        802.1q         trunking      99

Port      Vlans allowed on trunk
Gig0/1    10,20,99
Gig0/2    10,20,99

Port      Vlans allowed and active in management domain
Gig0/1    none
Gig0/2    none

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    none
Gig0/2    none

SW2#
```

SW3:

```
SW3>enable
SW3#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#int g0/2
SW3(config-if)#switchport mode trunk
SW3(config-if)#switchport trunk native vlan 99
SW3(config-if)#switchport trunk allowed vlan 10,20,99
SW3(config-if)#end
SW3#
%SYS-5-CONFIG_I: Configured from console by console

SW3#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Gig0/2    on        802.1q         trunking      99

Port      Vlans allowed on trunk
Gig0/2    10,20,99

Port      Vlans allowed and active in management domain
Gig0/2    none

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/2    none

SW3#
```

Task 92 – Verify VLANS

Trunking

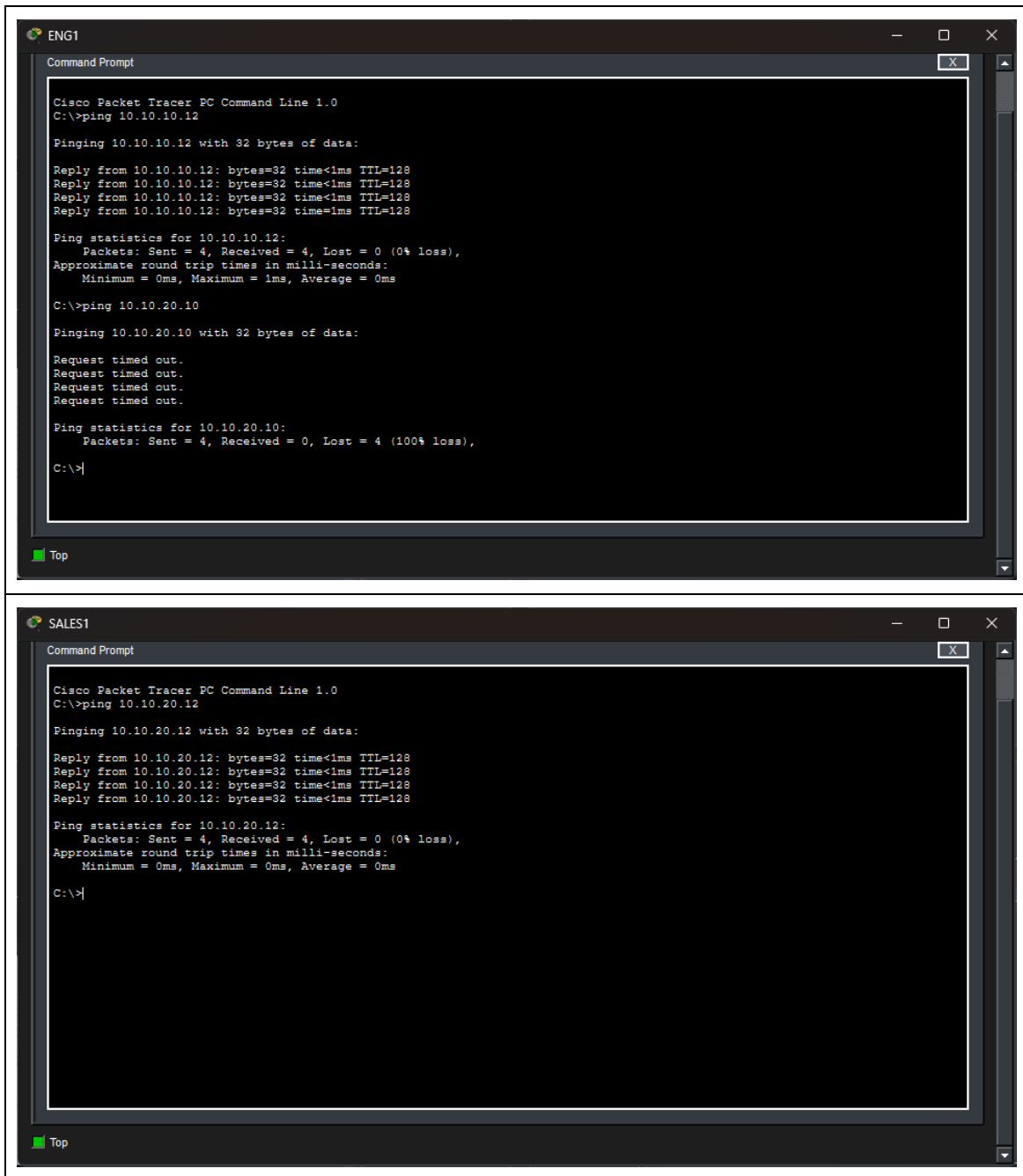
The image displays three screenshots of Cisco IOS Command Line Interface (CLI) windows for switches SW2, SW1, and SW3, showing the configuration and status of VLANs and trunking.

SW2: The CLI shows the configuration of VLANs 10 (ENG), 20 (SALES), 99 (NATIVE), 1002 (fddi-default), 1003 (token-ring-default), 1004 (fddinet-default), and 1005 (trnet-default). The trunking status is shown for Gig0/1 and Gig0/2, both configured as trunking ports with native VLAN 99. The trunking status is also shown for Gig0/1 and Gig0/2, both configured as trunking ports with native VLAN 99.

SW1: The CLI shows the configuration of VLANs 10 (ENG), 20 (SALES), 99 (NATIVE), 1002 (fddi-default), 1003 (token-ring-default), 1004 (fddinet-default), and 1005 (trnet-default). The trunking status is shown for Gig0/1, configured as a trunking port with native VLAN 99. The trunking status is also shown for Gig0/1, configured as a trunking port with native VLAN 99.

SW3: The CLI shows the configuration of VLANs 10 (ENG), 20 (SALES), 99 (NATIVE), 1002 (fddi-default), 1003 (token-ring-default), 1004 (fddinet-default), and 1005 (trnet-default). The trunking status is shown for Gig0/2, configured as a trunking port with native VLAN 99. The trunking status is also shown for Gig0/2, configured as a trunking port with native VLAN 99.

Pings



Task 93 — Physical Interfaces

The image displays two screenshots of Cisco IOS Command Line Interfaces (CLI) for routers R1 and SW2.

Router R1: The CLI shows the configuration of interface FastEthernet0/1. The output of the `show ip interface brief` command is as follows:

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.1	YES	manual	up	up
FastEthernet0/1	10.10.20.1	YES	manual	up	up
Vlan1	unassigned	YES	NVRAM	administratively down	down

The output of the `show ip route` command is also displayed, showing the routing table with various codes (L, C, D, N1, N2, E1, E2, I, P) and their corresponding network addresses and metrics.

Switch SW2: The CLI shows the configuration of interface FastEthernet0/2. The output of the `show vlan brief` command is as follows:

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16, Fa0/17, Fa0/18, Fa0/19, Fa0/20, Fa0/21, Fa0/22, Fa0/23, Fa0/24
10 ENG	active	Fa0/1
20 SALES	active	Fa0/2
99 NATIVE	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Task 94 — Router on a Stick

The top screenshot shows the Cisco IOS Command Line Interface for router R1. The configuration includes interfaces FastEthernet0/0, 0.10, 0.20, and 0.99. The status of each interface is as follows:

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	manual	up	up
FastEthernet0/0.10	10.10.10.1	YES	manual	up	up
FastEthernet0/0.20	10.10.20.1	YES	manual	up	up
FastEthernet0/0.99	unassigned	YES	unset	up	up
FastEthernet0/1	unassigned	YES	manual	administratively down	down
Vlan1	unassigned	YES	NVRAM	administratively down	down

The bottom screenshot shows a Windows Command Prompt on a host named ENG1. The user enters the command `C:\>ping 10.10.20.10`. The output shows that the first four ping attempts failed (Request timed out), resulting in a 100% loss. After a second attempt, the ping succeeds, showing a 0% loss.

```
C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Task 95 — SVIs

The top screenshot shows the configuration of SW2. The interface status is as follows:

Interface	IP Address	Status	Line Protocol
FastEthernet0/17	unassigned	YES NVRAM	down
FastEthernet0/18	unassigned	YES NVRAM	down
FastEthernet0/19	unassigned	YES NVRAM	down
FastEthernet0/20	unassigned	YES NVRAM	down
FastEthernet0/21	unassigned	YES NVRAM	down
FastEthernet0/22	unassigned	YES NVRAM	down
FastEthernet0/23	unassigned	YES NVRAM	down
FastEthernet0/24	unassigned	YES NVRAM	down
GigabitEthernet0/1	unassigned	YES NVRAM	up
GigabitEthernet0/2	unassigned	YES NVRAM	up
Vlan1	unassigned	YES NVRAM	administratively down
Vlan10	10.10.10.1	YES manual	up
Vlan20	10.10.20.1	YES manual	up

The route table for SW2 is as follows:

```
SW2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 2 subnets
C    10.10.10.0 is directly connected, Vlan10
C    10.10.20.0 is directly connected, Vlan20
```

The bottom screenshot shows the command-line interface of ENG1. The output of the ping command is as follows:

```
C:\>ping 10.10.20.10

Pinging 10.10.20.10 with 32 bytes of data:

Reply from 10.10.20.10: bytes=32 time<1ms TTL=127
Reply from 10.10.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The output of the traceroute command is as follows:

```
C:\>tracert 10.10.20.10

Tracing route to 10.10.20.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.10.10.1
  1  0 ms    0 ms    0 ms    10.10.20.10

Trace complete.
```

Reflection Questions

Task 96 – VLAN Scaling

Out of the 3 options. The best one for scalability in a large campus would be using a Layer 3 Switch with SVIs. The reason for this is because the routing for it occurs in Hardware at wire speed and sits inside the backplane of the switch. In other words, the traffic never leaves the switch meaning that trunking is not a bottleneck. The other benefit is that there is no external dependency, no extra hardware is required compared to the other options both of which require connecting an external router.